**Indian Institute of Technology, Kanpur**

**Department of SPASE Laboratory**

**Report Of Radio Astronomy Laboratory: SPA617**

# RF Components

***by***

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# 1.Aim

To experimentally characterize the remaining RF components covered in Part 2 of the RF Components laboratory, namely the power splitter/combiner and RF amplifier, including amplifier gain versus frequency, harmonic/nonlinear behaviour, the 1-dB compression point, and third-order intercept (TOI).

# 2. Apparatus Required

- RF signal generator(s)
- Spectrum analyser
- RF power splitter/combiner( ZAPD-2-252-54)
- RF amplifier with DC power supply
- RF power combiner for two-tone excitation
- 50 Ω terminations
- RF cables, adapters and attenuators

# 3. Theory

## 3.1 Power splitter

A power splitter divides an input signal between two or more output ports. For an ideal equal two-way splitter, each output is 3 dB below the input. A practical splitter has additional insertion loss due to conductor loss, dielectric loss, connector loss and imperfect matching.

For the two output ports:

$$\mathbf{Loss_1 = Pin - Pout_1} \qquad \textbf{and} \qquad \mathbf{Loss_2 = Pin - Pout_2}$$




*Figure 1. Relevant experimental arrangement and splitter/combiner.*

## 3.2 Power combiner

A power combiner performs the reverse operation of a splitter. The output depends on the frequencies, amplitudes and relative phases of the signals applied to its input ports. If signals are not phase-aligned, destructive interference can reduce the combined output. The handwritten observations specifically note that phase differences between sources can produce a lower output because of destructive interference; therefore, same-phase combining is preferred when combining coherent signals.

## 3.3 RF amplifier gain and harmonics

The small-signal power gain of an RF amplifier is defined as the difference between output and input powers:

$$\textbf{Gain} = \textbf{Pout} - \textbf{Pin} \quad \textbf{(dB)}$$

For a nonlinear amplifier, the output is not purely proportional to the input. A polynomial representation contains higher-order terms. These terms generate harmonics and other distortion products. As the input power is increased, the amplifier eventually departs from its small-signal behaviour and its gain begins to decrease.

### 3.4 1-dB compression point

The 1-dB compression point is the input power at which the amplifier gain has fallen by 1 dB from its small-signal gain. The corresponding output power is the output 1-dB compression point. A plot of Pout versus Pin shows the departure from the approximately linear region, while Gain versus Pin shows the reduction in gain directly.

$$\text{Gain} = \text{Pout} - \text{Pin}$$

$$\text{Gain at 1-dB compression} = \text{Small-signal gain} - 1 \text{ dB}$$

### 3.5 Third-order intercept point (TOI)

When two closely spaced tones f1 and f2 are applied to a nonlinear amplifier, third-order intermodulation products appear at 2f1−f2 and 2f2−f1. In the idealized TOI construction, the fundamental output increases with an approximate 1:1 slope with input power, whereas the third-order intermodulation output increases with an approximate 3:1 slope. The extrapolated intersection is the third-order intercept point.

$$\text{InterModulating } 3^{rd} \text{ order frequencies} = 2f_1 - f_2 \text{ and } 2f_2 - f_1$$

## 4. Procedures

### 4.1 Power splitter

1. Connect the splitter input to the signal generator.
2. Connect one output to the spectrum analyser and terminate the other output with a 50 Ω load.
3. Measure the input power using the reference connection and measure the output power.
4. Interchange the output connections and repeat the measurement.
5. Calculate the losses to both output ports and compare them with the ideal 3 dB division.

### 4.2 Power combiner

6. Apply signals to the two input ports of the power combiner and observe the combined output on the spectrum analyser.
7. Record the input powers and calculate their corresponding linear powers where required.
8. Add the powers in mW to determine the ideal total available power for incoherent/equal-frequency power addition.
9. Compare the calculated total power with the measured combined output and account for cable/connector loss.
10. For coherent signals, consider the relative phase because constructive or destructive interference changes the combined output.

### 4.3 Amplifier gain versus frequency

11. Power the RF amplifier using the specified DC supply.
12. Set a safe fixed RF input level and measure the amplifier input power.
13. Insert the amplifier and measure output power over the operating frequency range.
14. Calculate Gain = Pout − Pin and plot gain versus frequency.

### 4.4 1-dB compression point

15. Select a frequency within the amplifier operating band.

16. Begin at a low input power and record Pout.
17. Increase Pin in safe steps and record Pout at every step.
18. Calculate Gain = Pout − Pin for each point.
19. Plot Pout versus Pin and Gain versus Pin on the same figure using separate vertical axes.
20. Determine the input power at which the gain has fallen by 1 dB from the small-signal gain.

### 4.5 Third-order intercept

21. Set two closely spaced input frequencies f1 = 50 MHz and f2 = 60 MHz, as recorded in the experiment.
22. Apply the two tones using the RF power-combiner arrangement.
23. Identify the fundamentals at 50 MHz and 60 MHz and the third-order products at 40 MHz and 70 MHz.
24. Record the output powers at 40 MHz, 70 MHz, 50 MHz and 60 MHz for several input-power values.
25. Plot Pout1 at 40 MHz with Pout3 at 50 MHz on one graph, and Pout2 at 70 MHz with Pout4 at 60 MHz on a second graph.
26. Fit straight-line trends to the measured data and extrapolate the trends to their intersection.
27. Record the extrapolated intersection power as the empirical intercept obtained from the recorded data.

## 5. Observation Tables

### 5.1 Power splitter observations

| Quantity | Observed value |
|---|---|
| Observed at Port 2 at 10 MHz | -4.5 dBm |
| Port 1 at 50 MHz with 50 Ω termination | -5.36 dBm |
| Port 1 at 50 MHz without 50 Ω termination | -6.25 dBm |
| Observed at port 1 at 10Mhz | -4.5 dBm |
| Port 2 at 50 MHz with 50 Ω termination | -5.50 dBm |
| Port 2 at 50 MHz without 50 Ω termination | -6.30dBm |
| Wire loss measured as | -2.01 dBm and -2.06dBm |
| Actual Power Splitter readings | -5.36+2.01= -3.35dBm and -6.30+2.06= -4.24 |

As all instruments have internal impredance as 50Ω , so we need to have 50Ω termination to avoid the transmission loss due to impedance mismatch. When the end terminal port is terminated with 50Ω impedance the loss gets reduced as observed and after deducting wire loss we receive the actual power splitted.

| Frequency | 50 MHz |
|---|---|
| Ideal output from 0 dBm input | -3.00 dBm per port |
| Actual Port 1 output | -3.35 dBm |
| Actual Port 2 output | -3.44 dBm |
| Port 1 loss | 0.35 dB |
| Port 2 loss | 0.44 dB |
| Wire loss measured as | -2.01 dBm and -2.06dBm |

Hence the loss of splitter is only -0.35dBm and – 0.344 dBm which shows it has a good splitter.

### 5.2 Power combiner observations

| Quantity | Observed/calculated value |
|---|---|
| Input 1 frequency | 10 MHz |
| Input 1 power | +1.72 dBm |
| Input 2 frequency | 10 MHz |
| Input 2 power | +0.93 dBm |
| Calculated total input power | 4.34 dBm |
| Measured combined output | -1.50 dBm |
| Measured wire loss | -0.85 dBm |
| Recorded actual combined power | -0.65 dBm |

Due to different source inputs from different devices , there are some internal phase fluctuations present that leads to internal destructive interference due to relative opposite phase cancellations . Hence, we observe the experiment by connecting the two sources from the output of a power splitters to the input of another power splitter which acts as combiner that gives same phase signal sources to avoid phase losses.

## 5.3 Amplifier gain versus frequency

| Frequency (MHz) | Pout (dBm) | Recorded Gain (dB) |
|---|---|---|
| 10 | 4.09 | 24.09 |
| 50 | 3.95 | 23.95 |
| 100 | 3.66 | 23.66 |
| 150 | 3.51 | 23.51 |
| 200 | 3.17 | 23.17 |
| 250 | 3.18 | 23.18 |
| 300 | 2.94 | 23.94 |
| 350 | 2.85 | 22.85 |
| 400 | 2.65 | 22.65 |
| 450 | 2.63 | 22.63 |
| 500 | 2.13 | 22.13 |

### 5.4 1-dB compression observations

| Pin (dBm) | Pout (dBm) | Gain = Pout − Pin (dB) |
|---|---|---|
| -20.00 | 3.92 | 23.92 |
| -19.00 | 4.86 | 23.86 |
| -18.00 | 5.85 | 23.85 |
| -17.00 | 6.84 | 23.84 |
| -16.00 | 7.83 | 23.83 |
| -15.00 | 8.83 | 23.83 |
| -14.00 | 9.81 | 23.81 |
| -13.00 | 10.80 | 23.80 |
| -12.00 | 11.70 | 23.70 |
| -11.00 | 12.75 | 23.75 |
| -10.00 | 13.72 | 23.72 |
| -9.00 | 14.67 | 23.67 |
| -8.00 | 15.59 | 23.59 |
| -7.00 | 16.39 | 23.39 |
| -6.00 | 17.04 | 23.04 |
| -5.00 | 17.17 | 22.17 |
| -5.75 | 17.23 | 22.98 |
| -5.63 | 17.23 | 22.86 |

## 5.5 Third-order intercept observations

| Pin (dBm) | Pout1 at 40 MHz (dBm) | Pout2 at 70 MHz (dBm) | Pout3 at 50 MHz (dBm) | Pout4 at 60 MHz (dBm) |
|---|---|---|---|---|
| -14 | -43.38 | -45.16 | 6.55 | 6.58 |
| -13 | -40.10 | -40.16 | 7.63 | 7.56 |
| -12 | -36.80 | -36.80 | 8.51 | 8.53 |
| -11 | -32.72 | -32.81 | 9.46 | 9.48 |
| -10 | -27.46 | -27.42 | 10.37 | 10.39 |
| -9 | -20.63 | -20.47 | 11.19 | 11.21 |
| -8 | -15.23 | -15.25 | 11.90 | 11.92 |
| -7 | -10.90 | -10.96 | 12.49 | 12.51 |

# 6. Graphs

## 6.1 Amplifier gain versus frequency

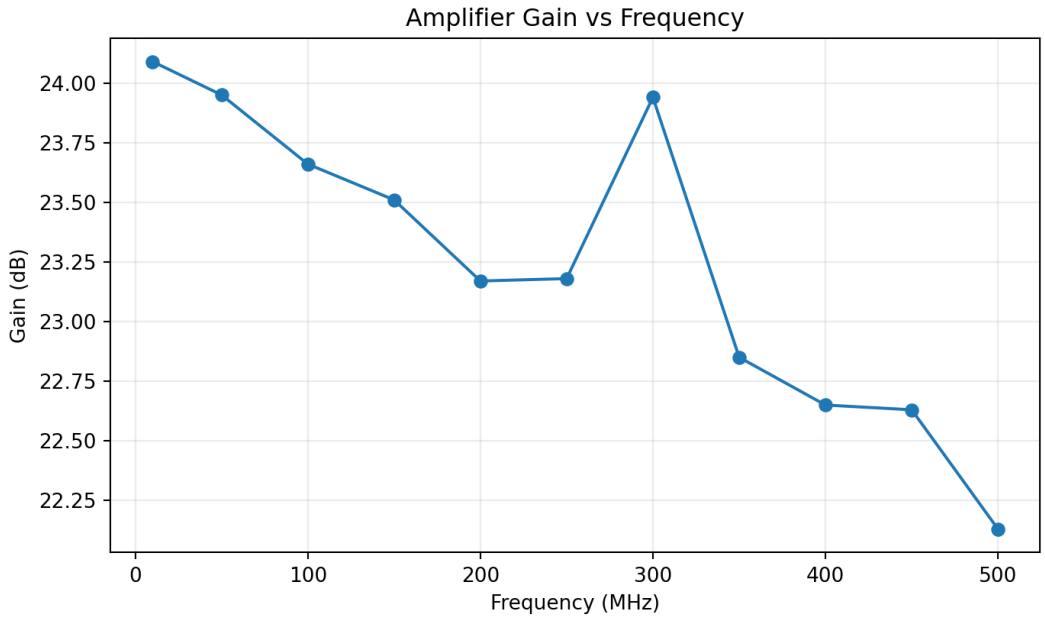



*Figure 5. Amplifier gain versus frequency using the recorded observations.*

The amplifier gain is highest at the lower frequencies and generally decreases as frequency increases. The recorded values range from 24.09 dB at 10 MHz to 22.13 dB at 500 MHz.

## 6.2 1-dB compression plot

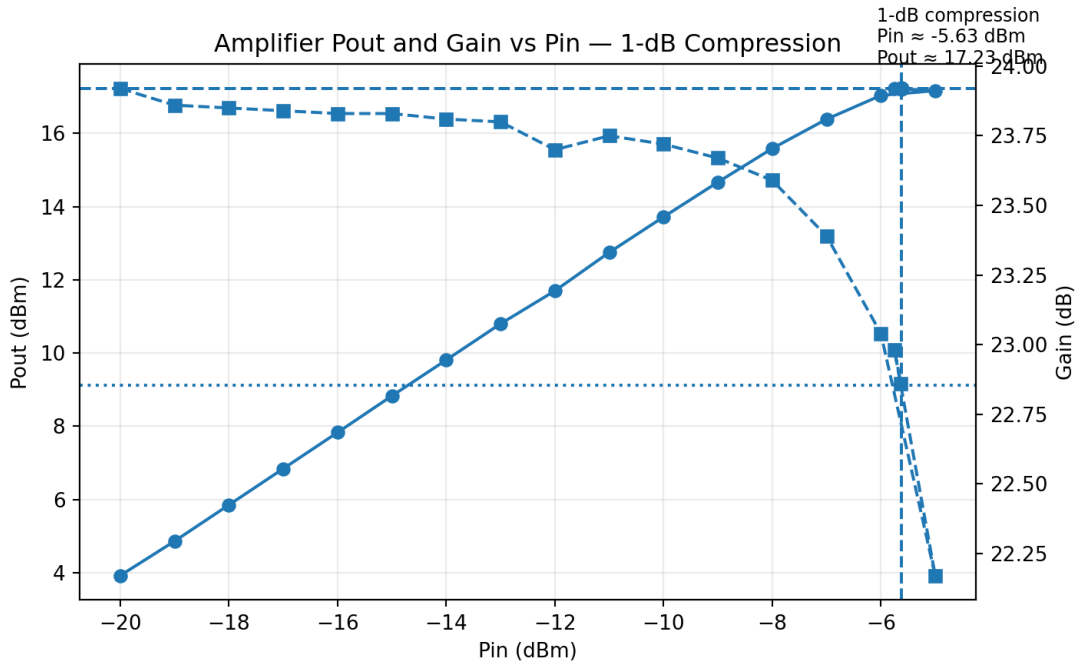



*Figure 6. Pout versus Pin and Gain versus Pin with the 1-dB compression point marked.*

The low-input gain is approximately 23.86 dB. Therefore, the 1-dB-compressed gain target is 22.86 dB. The recorded points near this level are Pin = −5.75 dBm (gain ≈ 22.98 dB) and Pin = −5.63 dBm (gain ≈ 22.86 dB). The experiment identifies −5.63 dBm as the 1-dB-compensated input power, with Pout ≈ 17.23 dBm.

## 6.3 Third-order intercept: 40 MHz and 50 MHz

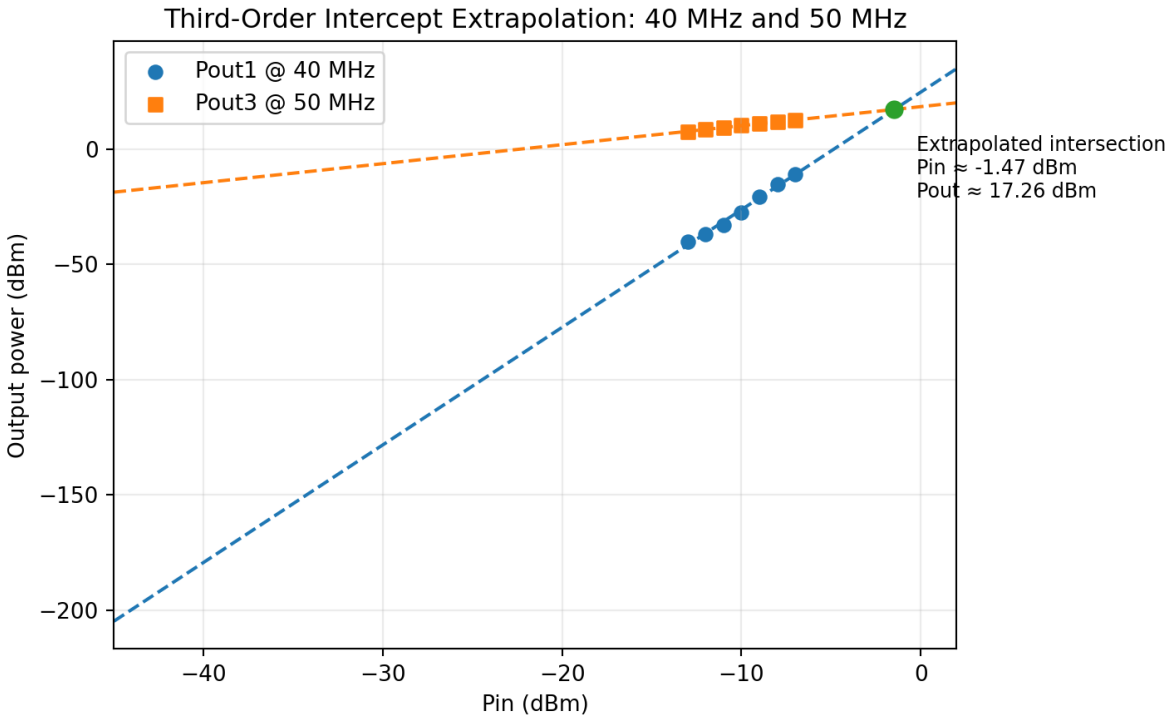



*Figure 7. Extrapolation of the recorded Pout1 (40 MHz) and Pout3 (50 MHz) trends.*

## 6.4 Third-order intercept: 70 MHz and 60 MHz

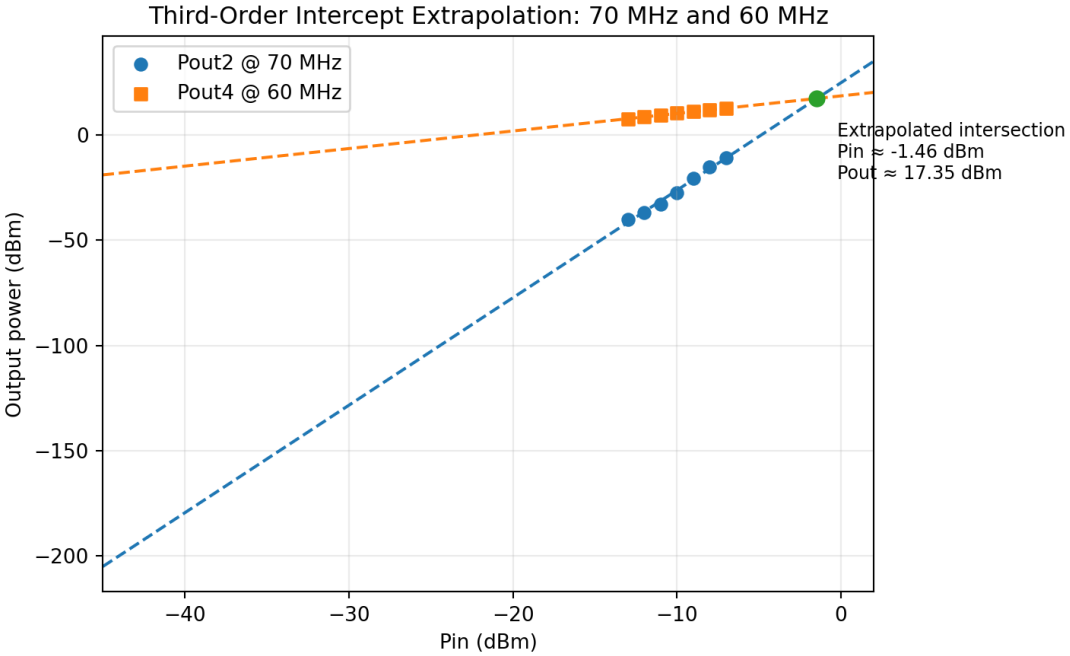



*Figure 8. Extrapolation of the recorded Pout2 (70 MHz) and Pout4 (60 MHz) trends.*

# 7. Calculations

## 7.1 Splitter loss

$$Ideal\ output = 0dBm - 3dB = -3dBm$$

From the recorded practical outputs:

$$Port1loss = (-3.00) - (-3.35) = 0.35dB$$

$$Port2loss = (-3.00) - (-3.44) = 0.44dB$$

The losses are small and unequal, which is consistent with a practical splitter rather than an ideal lossless device.

## 7.2 Power-combiner power calculation

For the two 10 MHz inputs, convert dBm to mW using:

$$P(mW) = 10^{(P(dBm)/10)}$$

For +1.72 dBm: $P_1 \approx 1.486$ mW. For +0.93 dBm: $P_2 \approx 1.239$ mW. Therefore total calculated power ≈ 2.725 mW = 4.35 dBm, consistent with value of 4.34 dBm.

The measured combined output is much lower than this ideal arithmetic sum. The  record attributes the difference to cable/wire loss and, importantly, to phase effects. For coherent signals, destructive interference can reduce the measured combined output when the two inputs are not phase-aligned.

## 7.3 Amplifier gain

$$\text{Gain} = \text{Pout} - \text{Pin}$$

For example, at 10 MHz the recorded output is 4.09 dBm. With the recorded nominal input of −20 dBm, the tabulated gain is 24.09 dB. The remaining frequency points are evaluated in the same manner and are plotted in Figure 5.

## 7.4 1-dB compression point calculation

$$\text{Small-signal gain} \approx \text{average of first six points} = 23.86\ \text{dB}$$

$$\text{1-dB compression target} = 23.86 - 1 = 22.86\ \text{dB}$$

At Pin = −5.75 dBm, Gain ≈ 22.98 dB; at Pin = −5.63 dBm, Gain ≈ 22.86 dB. Interpolating to 22.86 dB gives Pin ≈ -5.63 dBm. The corresponding recorded Pout is approximately 17.23 dBm. This agrees with the handwritten conclusion that the 1-dB compression occurs at approximately −5.63 dBm input.

## 7.5 Third-order intercept extrapolation

The manual describes the ideal construction using approximately 1:1 slope for the fundamental and 3:1 slope for the third-order product. The supplied measured data, however, do not follow those ideal slopes: a direct linear fit to the recorded points gives substantially different slopes. To avoid silently changing the observations, the graphs use straight-line regression through the actual recorded data and extend those lines until their measured-trend intersection.

For the 40 MHz / 50 MHz pair, the fitted equations are:

$$Pout1(40MHz) \approx 5.1011Pin + 24.7479$$

$$Pout3(50MHz) \approx 0.8246Pin + 18.4679$$

At the meeting point, the two output powers are equal. Therefore:

$$5.1011Pin + 24.7479 = 0.8246Pin + 18.4679$$

$$Pin \approx -1.47dBm, Pout \approx 17.26dBm$$

As an example of how the extrapolated points are generated, for any chosen Pin value x, substitute x into the fitted line. For example, at Pin = 0 dBm the first fitted line gives Pout1 ≈ 24.75 dBm and the second gives Pout3 ≈ 18.47 dBm. Repeating this for increasing Pin values produces the extended straight lines shown in Figure 7 until they intersect.

For the 70 MHz / 60 MHz pair:

$$Pout2(70MHz) \approx 5.1086Pin + 24.8186$$

$$Pout4(60MHz) \approx 0.8343Pin + 18.5714$$

$$Pin \approx -1.46dBm,$$

$$Pout \approx 17.35dBm$$

The two empirical intersections are approximately 17.26 dBm and 17.35 dBm. Their average is approximately 17.30 dBm., this value as an empirical extrapolated intersection of the recorded trends rather than as a strict ideal TOI.

# 8. Result

The completed quantities from Part 2 are summarized below, following the result-table structure of the laboratory manual.

| **Quantity** | **Measured / calculated value** |
|---|---|
| Splitter Port 1 output power | -3.35 dBm at 50 MHz |
| Splitter Port 2 output power | -3.44 dBm at 50 MHz |
| Splitter Port 1 loss | 0.35 dB |
| Splitter Port 2 loss | 0.44 dB |
| Combiner calculated total input power | 4.34 dBm |
| Combiner recorded actual combined power | -0.65 dBm |
| Amplifier input power used for gain-vs-frequency table | Nominal −20 dBm (handwritten calculation basis) |
| Amplifier output at 10 MHz | 4.09 dBm |
| Amplifier gain at 10 MHz | 24.09 dB |
| Amplifier output at 500 MHz | 2.13 dBm |
| Amplifier gain at 500 MHz | 22.13 dB |
| 1-dB compression input power | ≈ −5.63 dBm (recorded conclusion; calculated target ≈ −5.62 dBm) |
| 1-dB compression output power | ≈ 17.23 dBm |
| 1-dB compressed gain | ≈ 22.86 dB |
| Empirical TOI intersection, 40/50 MHz pair | ≈ 17.26 dBm |
| Empirical TOI intersection, 70/60 MHz pair | ≈ 17.35 dBm |
| Average empirical extrapolated TOI | ≈ 17.30 dBm |

# 9. Discussion

- The two-way splitter produces approximately −3 dBm at each output for a 0 dBm input in the ideal case. The measured values of −3.35 dBm and −3.44 dBm show only a small additional loss, indicating reasonably good splitter performance.

- The two splitter outputs are not exactly equal. This is expected in a practical component because of insertion loss, mismatch and unequal path losses.
- For the combiner, the linear-power calculation gives approximately 4.34 dBm from the two recorded 10 MHz input powers. The much smaller measured combined output is explained in the handwritten record by wire/cable loss and by relative phase differences. When two coherent signals are out of phase, destructive interference reduces the resultant power.
- The amplifier gain is highest at the low-frequency end and generally falls with frequency. This indicates that the amplifier does not maintain a perfectly flat gain response over the measured 10–500 MHz range.
- As Pin increases, Pout initially rises almost linearly and the gain remains nearly constant. Near the high-input region, Pout begins to flatten and the gain decreases. This is the expected onset of nonlinear compression and is the physical reason for the 1-dB compression point.
- The measured 1-dB compression point occurs near −5.63 dBm input, where the gain is approximately 1 dB below the small-signal value and Pout is approximately 17.23 dBm.
- The third-order experiment shows the expected presence of output components at 40 MHz and 70 MHz for f1 = 50 MHz and f2 = 60 MHz, because 2f1−f2 = 40 MHz and 2f2−f1 = 70 MHz.
- The laboratory manual describes ideal TOI extrapolation with approximately 1:1 fundamental and 3:1 third-order slopes. The recorded data in the supplied notebook do not exhibit those ideal slopes. Therefore, the report does not alter the observations to force ideal behaviour; instead, it shows empirical straight-line extrapolation of the recorded points and explicitly identifies the resulting intersections as empirical.
- Possible causes of deviation include measurement uncertainty, spectrum-analyser level accuracy, cable losses, nonlinear response over the chosen input range, source phase/amplitude mismatch, and the fact that the selected measured points may not lie entirely in the small-signal and third-order-dominated regions required for an ideal TOI fit.

# 10. Conclusion

The Part 2 RF-components experiment was completed for the power splitter/combiner and RF amplifier. The splitter showed approximately 0.35–0.44 dB additional loss beyond the ideal 3 dB division. The combiner measurements demonstrated the importance of cable loss and relative phase in practical power combining. The amplifier gain decreased from approximately 24.09 dB at 10 MHz to 22.13 dB at 500 MHz. The 1-dB compression point was obtained at approximately -5.63 dBm input, consistent with the value of −5.63 dBm. The third-order data were extrapolated as instructed; the recorded trends give empirical intersections near 17.26 dBm and 17.35 dBm, with an average of approximately 17.30 dBm. The experiment provided practical understanding of the behaviour and characterization of important RF components and systems, namely the power splitter/combiner and RF amplifier. The experiments demonstrated how RF power is divided and combined, how amplifier gain varies with frequency, how an amplifier approaches nonlinear operation, and how nonlinear distortion can be quantified using the 1-dB compression point and third-order intercept point (TOI). The concepts investigated in this experiment have direct applications in real RF and microwave systems.

Some applications of the system:

1. **Radio communication systems** – RF amplifiers are used to increase weak received signals and to provide sufficient power for transmission. The gain, bandwidth, compression point and intermodulation performance determine the quality of the communication system.
2. **Mobile and wireless communication** – Base stations, Wi-Fi equipment, satellite terminals and other wireless systems use RF amplifiers, filters, splitters and combiners. Good linearity is essential because intermodulation products can interfere with neighbouring communication channels.
3. **Radio astronomy** – Radio telescopes receive extremely weak electromagnetic signals from astonomical objects such as pulsars, galaxies, quasars, supernova remnants and molecular clouds.